



WEST BENGAL STATE UNIVERSITY
B.Sc. Major 2nd Semester Examination, 2024

CMSDSC202P-COMPUTER SCIENCE (MAJOR)
(PRACTICAL)

SET-I

Time Allotted: 2 Hours

Full Marks: 50

*The figures in the margin indicate full marks.
Candidates should answer in their own words and adhere to the word limit as practicable.
All symbols are of usual significance.*

Answer any one question from the following

[Programs should be written in C++]

1. Write a program using recursion
(i) to display Fibonacci series upto n numbers. (n should be input by the user)
(ii) to compute the factors of a given number.
2. Write a program to search an element from a list. Give user the option to perform Linear search.
3. Write a program to perform Stack operations using Linked List implementation.
4. Write a program to implement Linked List. Include functions for insertion, deletion and display.
5. Write a program to reverse a given string using stack.
6. Write a program to perform Queues operations.
7. Write a program to sort a list of elements. Give user the option to perform sorting using Insertion sort.
8. Write a program to implement Double Linked List. Include functions for insertion, deletion and display.
9. Write a program to scan two polynomials using Array and add two polynomials.
10. Write a program to implement Lower Triangular Matrix using a one-dimensional array.
11. Write a program to transform an infix expression to postfix expression.
12. Write a program to implement Binary Search in a sorted array.
13. Add two polynomials that are represented in Linked List.
14. Write a program to create a Binary Search Tree and include Insertion operation in the tree.

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Candidates should answer in their own words and adhere to the word limit as practicable.

All symbols are of usual significance.

Answer any *one* question from the following

[Programs should be written in C++]

1. Write a program to implement Circular Linked List. Include functions for insertion, deletion and display.
2. Write a program to search an element from a list. Give user the option to perform Binary search. Use Template functions.
3. Write a program to sort a list of elements. Give user the option to perform sorting using Bubble sort.
4. Write a program using recursion to calculate GCD of two numbers.
5. Write a program to implement Linked List. Include functions for insertion, display and reverse the list.
6. Write a program to reverse the order of the elements in the stack using additional stack.
7. Write a program to perform Stack operations using Array implementation.
8. Write a program to sort a list of elements. Give user the option to perform sorting using Insertion sort. Use Template functions.
9. Write a program to transform an infix expression to postfix expression.
10. Write a program to convert the Sparse Matrix into non-zero form and vice-versa.
11. Write a program to implement a queue and perform various functions on it.
12. Write a program to implement Circular Linked List. Include function for insertion, display and reverse the list.
13. Write a program to add two polynomials using Linked List.
14. Write a program to create a Binary Search Tree and include following operations in tree:
 - (i) Insertion
 - (ii) Deletion.

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